From Noise to Diversity: Random Embedding Injection in LLM Reasoning

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Abstract

Recent *soft prompt* research has tried to improve reasoning by inserting trained vectors into LLM inputs, yet whether the gain comes from the *learned content* or from the *act of injection itself* has not been carefully separated. We study Random Soft Prompts (RSPs), which drop the training step entirely and append a freshly drawn sequence of random embedding vectors to the input. Each RSP vector is sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table; the sequence carries no learned content, and yet reaches accuracy comparable to optimized *soft prompts* on math reasoning benchmarks in several settings. The mechanism unfolds in two stages: because attention has to absorb a never-seen-before random position, the distribution over the first few generated tokens flattens and reasoning trajectories *branch*, and as generation continues this influence dilutes naturally so the response commits to a single completion. We show that during inference RSPs lift early-stage token diversity and, combined with temperature sampling, widen Pass@N, the probability that at least one out of N attempts is correct. Beyond inference, we carry the same effect into DAPO training and demonstrate practical gains. Our contributions are: (i) RSP isolates the simplest form of *soft prompt* — training-free, freshly resampled — providing a unified lens for the structural effect of injection that variants otherwise differing in training and form all share; (ii) a theoretical and empirical validation of the underlying mechanism; and (iii) an extension from inference to training. `https://github.com/rsp-anon-2026/RandomSoftPrompt`

1 Introduction

This work began from a surprising observation: even *untrained*, randomly drawn embeddings appended to LLM inputs lift reasoning accuracy and reshape early decoding in several settings — with no optimization at all. As LLMs scale, full fine-tuning is prohibitive, motivating parameter-efficient methods [Hu et al., 2022, Houlsby et al., 2019]. Among these, *soft prompts* insert learnable continuous embeddings into the input and steer behavior through attention [Li and Liang, 2021, Lester et al., 2021]; the paradigm is actively adopted for mathematical reasoning [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026]. These methods diverge in injection position, training objective, and prompt form, and prior work attributes their gains to the *learned content* of the optimized vectors. The act of injecting embeddings into the input, common to all variants, has remained outside the analysis. Even though theory shows trained prefixes only bias attention in a fixed direction [Petrov et al., 2024], the source of the effect itself remains unsettled.

To separate *learned content* from the *act of injection*, we use Random Soft Prompts (RSPs) as a control — continuous vectors drawn from an isotropic Gaussian fitted to the embedding table's mean and variance, with no learning. The control turns out to be non-neutral. Applying a chat template to a base model not fine-tuned for it degrades reasoning, yet appending RSPs recovers up to $+29$

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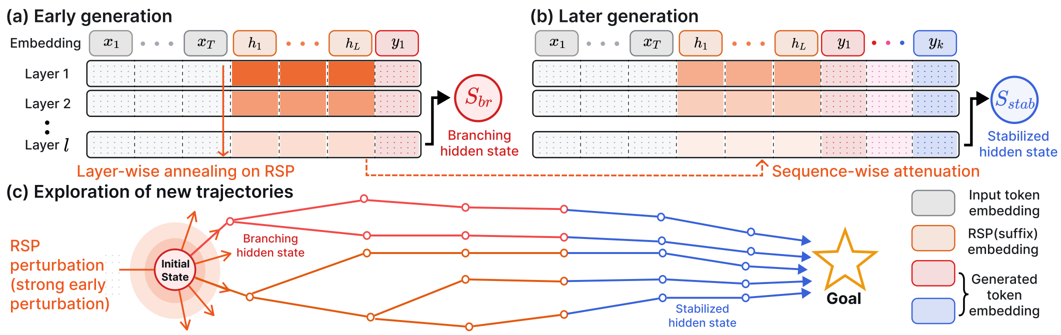

Figure 1: Conceptual overview of RSP-induced trajectory diversity. The hidden states shown are the final-layer outputs that drive the next-token distribution. (a) Early decoding: RSP perturbation reaches a *branching state* (red; formalized in §3.1), whose top-K differs from the no-RSP baseline and induces a diverse output distribution. (b) Later decoding: as the KV cache grows, the bound on RSP attention mass narrows (Theorem 1); branching events become rare and the hidden state (blue) commits to the branch already opened. (c) Across rollouts, the trajectories opened by early RSP perturbation accumulate into independent paths.

pp on Qwen2.5-Math-1.5B; if simple noise were merely shaking the model, accuracy should have worsened instead. The injection itself, independent of any learned content, alters model behavior.

We analyze training-free Random Soft Prompts (RSPs) on LLM reasoning. With each rollout receiving an *independent* draw, a single RSP injection *branches* hidden states into diverse trajectories early on, which then *stabilize* as decoding proceeds. Empirically, RSPs share the early-decoding entropy signature of optimized methods (LTPO, TTSV, SoftCoT) despite carrying no learned content; Pass@N accumulates only under *independent* resampling; and the same input-side diversity composes with DAPO [Yu et al., 2025] training. Our contributions are: (i) RSP isolates the simplest form of *soft prompt* — training-free and freshly resampled per rollout — providing a unified lens for the structural effect of injection across variants that otherwise differ in training objective and prompt form, while still reaching accuracy comparable to optimized methods in several settings; (ii) a theoretical and empirical account of the mechanism (attention-mediated branch opening, isotropic directional coverage, KV-cache attenuation); and (iii) DAPO extension showing independent RSP resampling composes with rollout-diversity training.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Prior *soft prompt* methods share a common structure: they inject continuous vectors that do not correspond to discrete vocabulary tokens into the model's representation stream and *optimize* them under task-specific objectives. As parameter-efficient alternatives to full fine-tuning, Prefix-Tuning [Li and Liang, 2021], Prompt Tuning [Lester et al., 2021], and P-tuning [Liu et al., 2024] reached near-fine-tuning performance using learnable continuous vectors, after which LLM-reasoning variants followed. TTSV [Kang et al., 2026] optimizes *prefix* embeddings at test time via trajectory entropy minimization; SoftCoT [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant model; LTPO [Ye et al., 2026] optimizes the *soft prompt* per problem to maximize its own confidence on that problem; COCONUT [Hao et al., 2025] feeds its hidden states back as input embeddings; MemGen [Zhang et al., 2026] uses embeddings as experience memory; and Pause tokens [Goyal et al., 2024] insert learnable tokens to provide additional computation steps.

These methods differ in position, training stage, and training target, yet share a common assumption: that the *learned representations* are the core driver of the effect. This paradigm faces several limitations. **(i)** Training is domain/benchmark-specific, requiring re-optimization for new settings and sensitive to seed/hyperparameter choices. **(ii)** The methods are brittle on out-of-distribution challenging tasks [Ye et al., 2026]. **(iii)** Evidence for the effect is largely empirical, with limited theoretical explanation beyond the training objective. Furthermore, soft prompting and prefix-tuning are theoretically restricted to biasing attention outputs in a fixed direction, without changing the

Table 1: Accuracy (%) across model configurations and benchmarks. Values in parentheses indicate the difference from the baseline. Full per-position breakdown is in Appendix A.

	MATH-500			GSM8K			AIME24		
Model	Baseline	RSP(*prefix*)	RSP(*suffix*)	Baseline	RSP(*prefix*)	RSP(*suffix*)	Baseline	RSP(*prefix*)	RSP(*suffix*)
Instruct Models									
LLaMA-3.1-8B-Inst	52.20	45.00 (−7.2)	49.40 (−2.8)	85.75	76.35 (−9.4)	86.73 (+1.0)	6.67	3.33 (−3.3)	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	81.40 (−1.8)	83.40 (+0.2)	95.45	94.92 (−0.5)	95.68 (+0.2)	13.33	13.33 (+0.0)	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	74.80 (+1.8)	74.20 (+1.2)	85.29	85.29 (+0.0)	84.69 (−0.6)	10.00	13.33 (+3.3)	20.00 (+10.0)
Base Models (Raw Text)									
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	55.60 (−16.6)	84.15	84.31 (+0.2)	54.13 (−30.0)	6.67	16.67 (+10.0)	13.33 (+6.7)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	54.60 (−6.8)	77.86	74.30 (−3.6)	58.61 (−19.3)	16.67	10.00 (−6.7)	3.33 (−13.3)
Base Models + ChatML (Format Mismatch)									
Qwen2.5-Math-7B	52.20	59.20 (+7.0)	70.40 (+18.2)	58.30	56.41 (−1.9)	75.97 (+17.7)	23.33	3.33 (−20.0)	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	51.00 (+16.6)	37.45	67.02 (+29.6)	50.64 (+13.2)	13.33	20.00 (+6.7)	13.33 (+0.0)

relative attention pattern among content tokens [Petrov et al., 2024]. Together, these findings cast doubt on whether the learned prompt vectors function as intended in prior work: they are suitable for *eliciting* or *combining* skills present in the pretrained model, but they cannot learn new capabilities.

Moreover, prior works have not separated the contribution of *injection itself* from that of *optimization*. In light of limitations (i)–(iii), this paper focuses on *injection itself*: we introduce *Random Soft Prompts* (RSPs) — vectors drawn from an isotropic Gaussian fitted to the pretrained embedding statistics, with the training stage entirely removed — and show that they produce output-distribution shifts and accuracy comparable to those of learned soft prompts, thereby disentangling the two contributions. Section 3 analyzes the mechanism theoretically; Section 4 validates it empirically.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at several stages and locations. During training, dropout [Srivastava et al., 2014] randomly deactivates *hidden activations*, and NEFTune [Jain et al., 2024] *adds* Gaussian noise to *input embeddings* during instruction fine-tuning to improve instruction following. At inference time, randomized smoothing [Cohen et al., 2019] injects Gaussian noise into *raw inputs* to obtain certified robustness, and SmoothLLM [Robey et al., 2025] perturbs prompts at the *discrete character* level to defend against jailbreaking. In RL, NoisyNet [Fortunato et al., 2018] *adds* learnable, trained noise to *network weights* to aid exploration. Rather than *adding* noise to embeddings, RSP *appends* it, and differs in two respects: **(i)** unlike NoisyNet, it is entirely training-free, with no model fine-tuning and no learned noise parameters; **(ii)** it appends within a single forward pass, eliminating the multi-pass aggregation that robustness methods require.

3 Random Soft Prompts and Why They Work

3.1 Random Soft Prompt

This section starts from an empirical observation. Random embedding vectors appended to LLM inputs lift reasoning accuracy across several models and benchmarks (Table 1) with no training; the gain holds across injection positions (*prefix*, *suffix*) (§4.2), traveling with the act of injection rather than a particular configuration. To explain why, we track one quantity — the *attention mass* w that the model places on RSP tokens. Early in decoding w is large enough to open a different decoding branch (§3.2); as the KV cache grows, the upper bound on w narrows automatically and the model commits to the branch already opened (§3.3) — an implicit explore-then-exploit. Independent resampling per rollout opens a fresh branch each time (proofs in Appendix D).

To isolate the act of injection from learned content while keeping the magnitude comparable to real tokens and not privileging any direction in embedding space, we draw RSP from an isotropic Gaussian fitted to the entrywise statistics of the pretrained embedding table. Let $\mathbf{W}_E \in \mathbb{R}^{V\times d}$ denote the pretrained token-embedding matrix, with V the vocabulary size and d the hidden dimension. Write its entrywise statistics as $\mu_E := \frac{1}{V\cdot d}\sum_{v,k}[\mathbf{W}_E]_{v,k}$ (the mean over all $V\cdot d$ entries) and $\sigma_E :=$ $\mathrm{std}(\mathbf{W}_E)$. An RSP of length L is a sequence of continuous vectors $\mathbf{H} = [h_1,\ldots,h_L] \in \mathbb{R}^{L\times d}$

drawn independently from

$$h_j \;\sim\; \mathcal{N}(\mu_E \mathbf{1}_d,\; \sigma_E^2 \mathbf{I}_d), \quad j = 1, \ldots, L. \tag{1}$$

The centered form $\bar{h}_j := h_j - \mu_E \mathbf{1}_d \sim \mathcal{N}(0, \sigma_E^2 \mathbf{I}_d)$ is a zero-mean isotropic Gaussian. Letting $\mathbf{X} \in \mathbb{R}^{T \times d}$ denote the input token-embedding sequence (the T rows of $\mathbf{W}_E$ corresponding to the prompt tokens), the main text uses the *suffix* form $[\mathbf{X}; \mathbf{H}] \in \mathbb{R}^{(T+L) \times d}$ as the default; other positions (*prefix* $[\mathbf{H}; \mathbf{X}]$, *infix* ($\mathbf{H}$ inserted within $\mathbf{X}$) [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2026]) are reported as ablations in §4.2 and Appendix A. RSP is not a learnable parameter, so it receives no gradient and is freshly drawn for each rollout.

Call decoding step t a *branching event* and the last-layer hidden state $\mathbf{s}^{(t)}$ a *branching state* when the LM-head top-K differs from that of the no-RSP baseline $\bar{\mathbf{s}}^{(t)}$ (Figure 1, red).[1]

3.2 Exploration: RSP strengthens early-stage exploration

Inserting a random prompt concentrates attention on it, and that concentration amplifies exploration. The effect of a single injection on the hidden state raises two questions — *how much* and *where* — decided respectively by the scalar w inside one attention head and by the distribution of $\mathbf{H}$. The isotropic Gaussian addresses both at once. Both questions trace back to the branching condition of §3.1: for a single RSP draw to shift top-K, its perturbation of the head output must be large enough to propagate through the layers (*how much*) and along directions that affect logit ordering (*where*).

Consider one attention head in self-attention layer ℓ. At query position i, the query attends $n \geq 1$ unmasked real tokens together with $L \geq 1$ RSP tokens, all attention logits finite. Write the attention weights as $\alpha_{ij}^{(\ell)}$ and the value vectors on the real and RSP sides as $\mathbf{v}_j^{(\ell)}, \mathbf{v}_{r_j}^{(\ell)}$. Defining the total attention mass on random tokens $w_{r,i}^{(\ell)} := \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)}$, the real-token mass $1 - w_{r,i}^{(\ell)}$ is positive and the head output $\mathbf{o}_i^{(\ell)}$ splits *exactly* into a renormalized real-token term $\tilde{\mathbf{o}}_i^{(\ell)}$ and an RSP-induced contribution $\boldsymbol{\eta}_i^{(\ell)}$:

$$\mathbf{o}_i^{(\ell)} \;=\; (1 - w_{r,i}^{(\ell)})\, \tilde{\mathbf{o}}_i^{(\ell)} \;+\; \boldsymbol{\eta}_i^{(\ell)}, \quad \tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^{n} \tfrac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}}\, \mathbf{v}_j^{(\ell)}, \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}. \tag{2}$$

Derivation of Eq. (2). Write the head output as $\sum_{j=1}^{n} \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}$, then factor $1 - w_{r,i}^{(\ell)} = \sum_{j=1}^{n} \alpha_{ij}^{(\ell)} > 0$ out of the real-token sum. No approximation beyond the softmax-normalization identity is used. $\square$

This single quantity $w_{r,i}^{(\ell)}$ controls both the attenuation ratio $1 - w_{r,i}^{(\ell)}$ on the real signal and the upper bound on the random contribution $\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_j \|\mathbf{v}_{r_j}^{(\ell)}\|$. The *magnitude* of RSP-induced variation thus reduces to one number in $[0, 1]$. Early in decoding the KV cache is short and this value is non-negligible, so a single RSP draw can perturb next-token decisions.

The scalar w controls *how much*; *where* is decided by the distribution of $\mathbf{H}$. For the local logit argument, isolate one RSP position and let $\bar{h} \in \mathbb{R}^d$ denote the i.i.d. marginal of any centered vector $\bar{h}_j$ in Eq. (1), with the other centered RSP positions fixed at zero. Let $z_a(\bar{h})$ be the under-RSP output logit at vocab token a and write the vocab-logit gap $\Delta_{ab}(\bar{h}) := z_a(\bar{h}) - z_b(\bar{h})$ ($\bar{h} = 0$ is the centered mean, not the no-RSP state; Appendix C). The transformer's logit map is non-linear in $\bar{h}$, but a first-order Taylor expansion around $\bar{h} = 0$ gives the local surrogate $\Delta_{ab}(\bar{h}) \approx \Delta_{ab}(0) + b_{ab}^\top \bar{h}$, with $b_{ab} := \nabla_{\bar{h}} (z_a - z_b)|_{\bar{h}=0} \in \mathbb{R}^d$ the gradient direction along which vocab tokens a, b swap rank (not unit-norm in general). Because RSP is training-free, we cannot know in advance which b_{ab} opens a useful branch, so the distribution must avoid systematically under-perturbing any direction. A deterministic injection pins to one direction; vocabulary sampling inherits the embedding table's anisotropy. The remaining design — maximize the worst-case directional variance at fixed budget — is uniquely solved by isotropy.

[1] Italic h_j: RSP input space; bold $\mathbf{s}^{(t)}$: hidden state. K denotes the cardinality in top-K analyses; nucleus sampling uses a separate threshold p. t for decoding step, ℓ for self-attention layer.

Proposition 1 (Maximin directional coverage)**.** *Let $\mathcal{D}_\rho$ denote the family of zero-mean distributions on $\mathbb{R}^d$ whose covariance satisfies $\mathrm{tr}(\Sigma_D) \leq \rho^2$. For each $D \in \mathcal{D}_\rho$, $\min_{\|u\|=1} \mathrm{Var}_{h\sim D}(u^\top h) = \lambda_{\min}(\Sigma_D)$, and $\sup_{D\in\mathcal{D}_\rho} \lambda_{\min}(\Sigma_D) = \rho^2/d$, attained iff $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.*

This is a *design criterion* for the prompt law, not a guarantee of correctness; correctness enters through the task-side $p_{\min}(x)$ assumption of §3.3 and independent resampling.[2]

Gaussian RSP attains the equality with $\Sigma = \sigma_E^2 \mathbf{I}_d$ ($\rho^2 = d\sigma_E^2$) and adds *full support* on $\mathbb{R}^d$ (Appendix C). Isotropy excludes no direction; full support gives positive probability to the open branching-event region (§3.1) whenever it is non-empty (Appendix D.3.3). Monotone temperature preserves ranking and cannot reach this region, so RSP's *rank-changing* branching follows from the two properties together, splitting the hidden state into multiple branching states early in reasoning.

3.3 Annealing: KV cache growth dilutes RSP attention

The early branching stabilizes as decoding proceeds: the upper bound on $w_{r,i}^{(\ell)}$ narrows along the sequence axis with autoregressive cache growth. Each step adds one real-token term while RSP terms stay fixed; at frozen attention-logit gap, this asymmetry yields the following bound.

Theorem 1 (Attention-mass decay under KV cache growth)**.** *Consider a single attention head with per-head key dimension d_{head}, where query $\mathbf{q}_i$ at position i attends $n \geq 1$ real keys $\mathbf{k}_j$ and $L \geq 1$ RSP keys $\mathbf{k}_{r_{j'}}$ under finite logits. Define the pre-scale* attention-logit *gap $\Delta_i := \min_{j\in[n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j'\in[L]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ (distinct from the vocab-logit gap Δ_{ab} of §3.2). In scaled dot-product attention,*

$$w_{r,i} \;\leq\; \frac{L}{L + n\,\exp(\Delta_i/\sqrt{d_{head}})},$$

and the right-hand side is strictly decreasing in n and tends to 0 at fixed L and Δ_i.

Proof. Let $s_j := \mathbf{q}_i^\top \mathbf{k}_j/\sqrt{d_{\text{head}}}$, $s_{r_{j'}} := \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}/\sqrt{d_{\text{head}}}$, $s_{r,\max} := \max_{j'} s_{r_{j'}}$, and $R := \sum_{j'=1}^{L} \exp(s_{r_{j'}})$. The gap definition gives $s_j \geq \Delta_i/\sqrt{d_{\text{head}}} + s_{r,\max}$ for every real j, and $\exp(s_{r,\max}) \geq R/L$ holds because the max dominates the average; combining the two and summing over the n real keys yields $\sum_{j=1}^{n} \exp(s_j) \geq (n/L)\exp(\Delta_i/\sqrt{d_{\text{head}}})\, R$. Substituting into $w_{r,i} = R/(\sum_j \exp(s_j) + R)$ gives

$$w_{r,i} \;\leq\; \frac{R}{(n/L)\exp(\Delta_i/\sqrt{d_{\text{head}}})\,R + R} \;=\; \frac{L}{L + n\,\exp(\Delta_i/\sqrt{d_{\text{head}}})}.$$

The derivative of the right-hand side in n is $-L\exp(\Delta_i/\sqrt{d_{\text{head}}})/(L + n\exp(\Delta_i/\sqrt{d_{\text{head}}}))^2 < 0$, so the bound is strictly decreasing and tends to 0 as $n \to \infty$. $\square$

Applied at each layer ℓ to $w_{r,i}^{(\ell)}$ with gap $\Delta_i^{(\ell)}$, L caps the maximum influence while n grows each step. The theorem guarantees only sequence-axis attenuation at fixed (layer, query, gap); the gap may sharpen at deeper layers, as suggested empirically by Figure 2, but that effect is outside the theorem's scope. The accurate reading is "the attainable upper bound at a fixed gap decreases monotonically." Because Eq. (2) ties the random contribution to the same scalar, branching-event reachability inherits this envelope — as the cache grows, events become rare and the response commits to the branch already opened, an implicit explore-then-exploit. Figure 2 visualizes this pattern along both axes.

Lifting single-shot branching to task accuracy across N rollouts requires (M1) a positive lower bound $p_{\min}(x) > 0$ on the per-rollout probability that one execution reaches a correct trajectory and (M2) the N executions are mutually independent. Under both, $\Pr(\text{Pass@}N \text{ on } x) \geq 1 - (1 - p_{\min}(x))^N$ (Appendix D.3.4). §3.2's isotropy and full support make local branch opening reachable when the open region is nonempty, but they do not by themselves guarantee the task-side correctness condition (M1); *independent resampling* supplies (M2). Sharing one RSP breaks (M2) and removes the accumulation — a signature §4 verifies empirically.

[2]The maximin is stated in input space; isotropy is not claimed to survive transformer layers. Any logit direction pulls back via the model Jacobian to an input-space b, and isotropy guarantees only that no such pulled-back direction (in the role of u above) has zero projected variance.

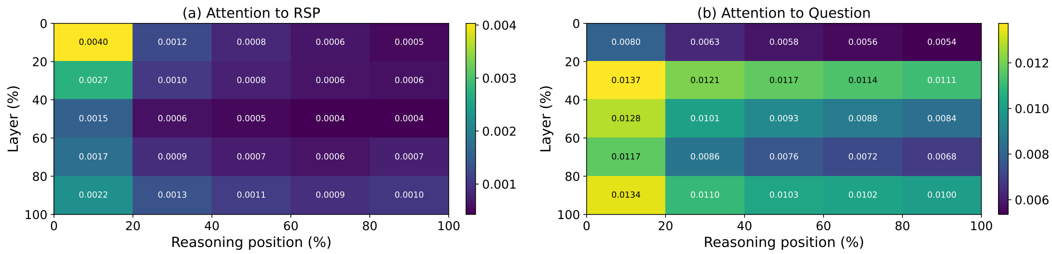

Figure 2: Per-token attention mass on Qwen2.5-Math-7B (*suffix*, 500 MATH-500 problems, independent RSP per problem). (a) RSP-token attention decreases along the reasoning-position axis, matching Theorem 1's KV cache-growth bound; the additional layer-axis decrease is an empirical phenomenon (deeper layers sharpen the real-vs-random gap) outside Theorem 1's scope. (b) Question-token attention stays comparatively uniform. Reported quantity: per-token mass of Appendix E, equal to $w_{r,i}^{(\ell)}/L$ averaged over heads and samples.

4 Empirical Validation

The mechanism of §3 runs from early *branching* through KV cache-growth narrowing to Pass@N accumulation under independent resampling. We verify this flow across three scenes. First, §4.2 tests whether RSPs preserve baseline accuracy and even recover it in some settings. Next, §4.3 examines whether the predicted early diversification and later stabilization appear in entropy and attention signals. §4.4 asks whether the Pass@N gain depends on *independence* of the RSP draw. Finally, §4.5 extends the same perturbation from inference to RL training, where rollout diversity is critical.

4.1 Experimental Setup

We evaluate on three mathematical reasoning benchmarks, MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across instruct models, base models, and base models with mismatched chat templates. Tables 1 and 2 use greedy decoding to isolate the effect of RSPs from sampling stochasticity; the diversity and Pass@N experiments in §4.4 use the sampling settings specified there. RSPs follow the definition of §3.1, using the default *suffix* position with a freshly drawn $\mathbf{H}$ per rollout. Since Theorem 1 identifies the RSP length L as the design parameter capping the maximum perturbation influence, we ablate $L \in \{10, 15, 20\}$ per model to identify an appropriate perturbation strength. Tables 1 and 2 report results at the per-model selected length. The mechanism analyses in §4.3–§4.4 use a fixed setting (L=10, *suffix*), which insulates them from this selection effect. Full experimental details and per-position results are in Appendix A.

4.2 How Does Untrained RSP Compare to Baselines & Optimized Soft Prompts?

We unpack the per-configuration patterns of Table 1 (presented in §3). On instruct models RSP stays within a few percentage points of the baseline. The most striking pattern is format-mismatch recovery: when a ChatML template is applied to a base model not trained for that format, *Suffix* recovers $+18.2/+17.7$ pp on Qwen2.5-Math-7B (MATH-500/GSM8K) and *prefix* recovers up to $+29.0/+29.6$ pp on Qwen2.5-Math-1.5B — if simple noise were responsible, accuracy should have dropped further. In the base raw-text setting the picture inverts: *suffix* incurs sharp losses ($-16.6/-30.0$ pp on Qwen2.5-Math-7B) while *prefix* changes accuracy less, suggesting base models treat EOS-adjacent signals as a derailment trigger that *prefix* avoids.

RSP's *direct accuracy gains* concentrate in misaligned-input recovery and stay within ±a few percentage points on well-aligned instruct models. The effects this paper centers on are the token-, trajectory-, and outcome-level diversity in §4.4, with accuracy recovery as a secondary outcome.

Compared with three optimized methods — TTSV, SoftCoT, LTPO — under a unified pipeline (Appendix A.2), RSP lands in the same accuracy band without any training, optimization, or task-specific tuning. It wins outright on several model–benchmark cells and shows meaningful gains even on the challenging AIME24 setting. What we want to highlight is the converse: a fixed, training-free

Table 2: Comparison with prior *soft prompt* methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	**84.20**	83.80	82.80	83.00
	GSM8K	95.45	**95.68**	95.30	95.00	94.84
	AIME24	13.33	16.67	**23.30**	16.67	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	75.20	75.20	**76.00**	72.40
	GSM8K	85.29	**85.75**	85.70	84.46	84.53
	AIME24	10.00	**20.00**	6.70	6.67	10.00

distribution reaching this range is the paper's central evidence that part of the *soft-prompt* effect comes less from learned content than from the random directional perturbation that injection induces — maximin coverage (§3.2) at a magnitude bounded by the KV cache (§3.3).

4.3 How Does RSP Affect the Output Distribution?

We next examine how RSP injection reshapes the output distribution. The setting is Qwen2.5-Math-1.5B-Instruct on MATH-500, and we measure per-token entropy, top-1 probability, and varentropy across generation steps. Figure 3 compares these metrics for the first 5% of generation steps across RSP variants and prior *soft prompt* methods (LTPO, SoftCoT, TTSV); full curves are in Appendix A.4.

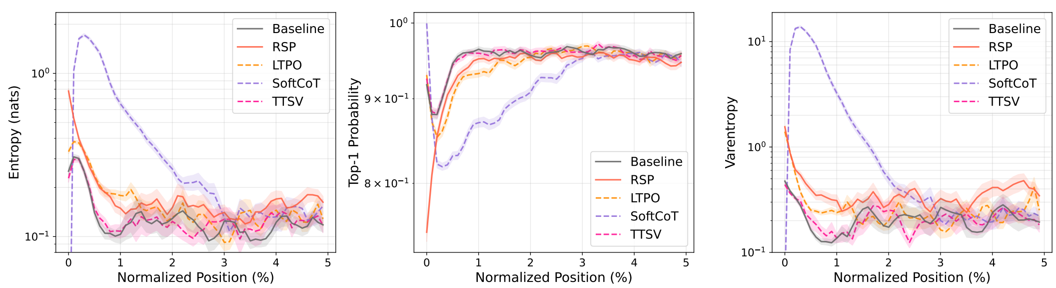

Figure 3: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior *soft prompt* methods (LTPO, SoftCoT, TTSV).

Latent prompt methods including RSP share the same signature: *a rise in early-generation entropy followed by convergence to the baseline later in generation*. A natural mechanistic explanation is the out-of-vocabulary (OOV) effect — when continuous embeddings outside the vocabulary enter the input, the model has to attend to a never-before-seen key position on top of its familiar token distribution, redistributing probability mass across multiple candidates within the early next-token distribution. All three RSP injection positions show early-stage entropy↑, top-1 probability↓, and varentropy↑, jointly consistent with multimodal distributions [Ahmed et al., 2026]. The change is confined to the early stage and all three metrics return to baseline in the middle and final portions, consistent with the *early influence with automatic KV cache-growth decay* pattern predicted in §3.3.

Prior methods trained under different objectives (LTPO, SoftCoT, TTSV) share this signature. Even TTSV, whose reward explicitly *lowers* mean reasoning-trajectory entropy, shows early-stage entropy comparable to the baseline. The entropy value itself is not a quality score — higher entropy is not always better; rather, the signature reveals a *shared mechanism* in which OOV embedding injection produces the same fingerprint independently of the optimization objective, supporting the hypothesis that the key driver of latent reasoning is the *act of injection*, not the learned content.

4.4 Does RSP Induce Trajectory Diversity?

Section 4.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. The three analyses converge on a directional picture: RSP's first-token perturbation propagates into semantically more diverse reasoning trajectories, which in turn yields greater output diversity at the task level. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, *suffix* injection, and $L = 10$.

Token-level: distribution beyond temperature. We quantify how much RSP shifts the first-token distribution — the critical forking point for reasoning trajectories [Wang et al., 2025]. Using the baseline at τ=1.0 as reference, we compare baselines at τ=2.0, 3.0 (16 samples per problem) against RSP at τ=1.0 (averaged over 16 seeds) on three metrics: Spearman rank correlation ρ to detect rank reorganization, the probability mass placed outside the baseline's top-10 to measure support expansion, and Jensen–Shannon divergence to quantify the overall magnitude of distributional change.

Table 3 shows the contrast: RSP's distributional shift falls between τ=2.0 and τ=3.0 in magnitude, but the mechanism differs — temperature is a monotone transform that preserves ranking ($\rho = 1$ at any τ), while RSP partially preserves but reranks ($\rho = 0.709$). The Pass@1 row exposes the cost: matching RSP's magnitude with temperature toward τ=3.0 collapses accuracy to 1.42%, whereas RSP preserves 73.30%. Thus, RSP produces a fundamentally different perturbation than temperature scaling. Moreover, RSP's accuracy variability across 16 seeds at τ=1.0 is $\pm$1.07, comparable to the $\pm$1.47 that temperature sampling produces across 16 samples at τ=2.0 (detailed definitions in Appendix B).

Table 3: First-token distribution metrics measured against the baseline at τ=1.0, comparing baselines at higher temperatures (τ=2.0, 3.0) with RSP at τ=1.0. RSP metrics are averaged over 16 random seeds; Acc reports mean $\pm$ std across 16 samples per problem.

Metric	τ=2.0	τ=3.0	**RSP**
Spearman ρ	1.000	1.000	**0.709**
Mass outside top-10	5.18%	29.11%	**21.86%**
JS divergence	0.060	0.218	**0.131**
Acc (%)	20.77 $\pm$ 1.47	1.42 $\pm$ 0.35	**73.30 $\pm$ 1.07**

Trajectory-level: semantic diversity. Token-level reranking raises a second question: do first-token perturbations diverge into distinct trajectories, or converge to similar ones? We sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under Baseline and RSP at $\tau = 1.0$. Each trajectory is encoded by BGE-M3 [Chen et al., 2024] into a dense semantic vector, and we compute three metrics: *pairwise cosine distance* (overall semantic spread), *inter-cluster distance* between centroids of DBSCAN [Ester et al., 1996], a density-based clustering algorithm (separation between distinct trajectory groups), and *intra-cluster distance* (coherence within groups). Table 4 reveals three concurrent patterns: pairwise distance rises by $1.4\times$, inter-cluster distance jumps by $5.4\times$, and intra-cluster distance is essentially unchanged — the trajectory set becomes more diverse and splits into more separated groups while preserving within-group coherence. RSP also yields larger pairwise distance than baseline on 56 of 64 problems.

Table 4: Semantic diversity metrics (64 problems, 300 trajectories per condition).

Metric	**Baseline**	**RSP**
Pairwise cos. dist.	0.0612	**0.0879**
Inter-cluster dist.	0.0183	**0.0982**
Intra-cluster dist.	0.0526	0.0528

Outcome-level: Pass@N scaling. Does this token- and trajectory-level diversity translate into task-level outcome diversity, measured by Pass@N [Chen et al., 2021] (the probability that at least one of N sampled solutions is correct)? We generate 16 samples per problem on MATH-500 and AIME24 and compare four conditions: *(i) Baseline* at $\tau = 1.0$; *(ii) RSP (single seed)* at $\tau = 1.0$, with one RSP shared across all samples; *(iii) RSP (indep. seed)* with greedy decoding, a fresh RSP per sample; and *(iv) RSP (indep. seed)* at $\tau = 1.0$, a fresh RSP per sample.

Condition *(iv)* achieves the highest Pass@N on both benchmarks, reaching 92.80% on MATH-500 and 36.67% on AIME24 at $N = 16$, while *(ii)* shows no improvement over the baseline — the diversity gain depends on varying the embeddings across samples rather than fixing a single perturbation. Pass@1 stays comparable to the baseline on MATH-500, and on the harder AIME24 *(iv)* even surpasses it, suggesting that independent RSPs combined with temperature sampling expand trajectory diversity without sacrificing single-sample accuracy.

4.5 Application: DAPO Training with RSP

Beyond inference (§4), does the same effect transfer to training? DAPO [Yu et al., 2025] is a recent GRPO [Shao et al., 2024] variant that preserves rollout diversity at the loss level through asymmetric clipping. We use it as a testbed to verify whether input-side branch opening composes with loss-side

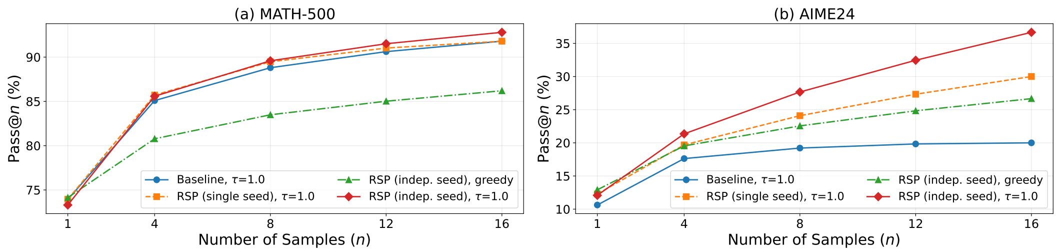

Figure 4: Pass@N scaling on (a) MATH-500 and (b) AIME24 with Qwen2.5-Math-1.5B-Instruct, 16 samples per problem. *Baseline*: temperature sampling only. *RSP (single seed)*: single RSP shared across samples combined with temperature. *RSP (indep. seed)*: a different RSP per sample, with or without temperature.

preservation to yield further gains. In DAPO+RSP, we incorporate the input-side perturbation of §4.4 into the RL rollout stage: each rollout is paired with an independently sampled suffix RSP. The full objective and implementation details are in Appendix F.

Table 5: Per-benchmark accuracy (%) at each method's peak step on Qwen2.5-Math-7B. **Bold** denotes the better of the two RL methods on each benchmark. Avg is the unweighted five-benchmark mean.

Method	GSM8K	MATH-500	College	Minerva	AIME24	Avg
Base	57.2	52.6	18.3	13.6	8.6	30.06
DAPO	86.3	78.2	41.4	37.5	22.6	53.20
DAPO + RSP	**87.7**	**78.6**	**42.2**	**39.7**	**23.6**	**54.36**

We implement DAPO on top of SimpleRL-Zoo [Zeng et al., 2025] and train Qwen2.5-Math-7B on a MATH Level-3–5 subset (~8.5K prompts): each step uses $G = 8$ rollouts per prompt at τ=1.0, batch 1,024, 4 PPO mini-batches of 256, for 100 steps. *DAPO* is compared against *DAPO + RSP*, where *suffix* RSP (L=20) is applied during rollouts only and evaluation uses no RSP. We evaluate on five math benchmarks (GSM8K [Cobbe et al., 2021], MATH-500 [Lightman et al., 2024], College Math, Minerva Math [Lewkowycz et al., 2022], AIME24) and report the unweighted mean; AIME24 uses Avg@32 at τ=1.0, the others greedy.

Figure 5 and Table 5 show two patterns: a late-stage advantage and a delayed early reward growth. DAPO + RSP reaches 54.36% at step 90 (vs DAPO's 53.20% peak, $+1.16$ pp) and widens the gap to $+3.10$ pp at step 100 with DAPO's post-peak decline delayed; earlier the curves cross around step 20. The patterns follow from the methods acting at separate stages of the rollout cycle: DAPO preserves diversity at the loss level over already-generated rollouts while RSP perturbs the hidden states that produce them (§3.2), exposing the policy to broader learning signals. Empirically, this is the exploration–exploitation dynamics of RL [Sutton and Barto, 2018] induced from the input side.

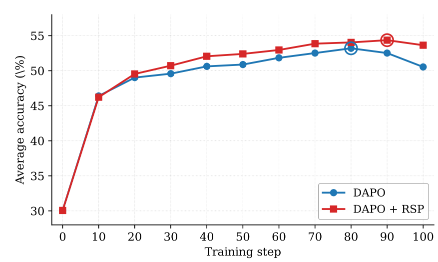

Figure 5: Five-benchmark average accuracy on Qwen2.5-Math-7B. DAPO + RSP reaches a higher peak (step 90) and stays stable through step 100.

5 Conclusion

We studied Random Soft Prompts (RSPs): training-free isotropic Gaussian vectors fitted to pretrained embedding statistics. The per-layer RSP contribution is controlled by an attention-mass scalar whose fixed-gap upper bound shrinks under KV cache growth, and isotropic resampling reaches top-K entry regions that rank-preserving temperature cannot (§3). Latent-prompt methods share the same entropy signature; the gain vanishes when one RSP is shared across samples (§4); DAPO + RSP yields a modest late-training advantage (§4.5). Because the signature emerges from purely random input-side perturbation, part of what trained *soft prompts* deliver appears to be a *structural by-product of injection*, and RSP composes orthogonally with sampling-based pipelines.

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A Experimental Setup and Full Accuracy Breakdown

All experiments are conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-Math-1.5B variants). The RSP token count L is selected per (model, benchmark) pair from $\{10, 15, 20\}$ (Table 7); the per-position breakdown across *prefix* ($[\mathbf{H}; \mathbf{X}]$), *infix* ($\mathbf{H}$ inserted within $\mathbf{X}$), and *suffix* ($[\mathbf{X}; \mathbf{H}]$) is in Table 6.

Table 6: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and <u>underline</u> denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
Instruct Models				
LLaMA-3.1-8B-Instruct	Baseline	**52.20**	85.75	<u>6.67</u>
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Infix	<u>49.40</u> (−2.8)	<u>85.97</u> (+0.2)	**10.00** (+3.3)
	Suffix	<u>49.40</u> (−2.8)	**86.73** (+1.0)	**10.00** (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	<u>13.33</u>
	Prefix	81.40 (−1.8)	94.92 (−0.5)	<u>13.33</u> (+0.0)
	Infix	**84.20** (+1.0)	<u>95.60</u> (+0.2)	**16.67** (+3.3)
	Suffix	<u>83.40</u> (+0.2)	**95.68** (+0.2)	**16.67** (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	<u>85.29</u>	10.00
	Prefix	<u>74.80</u> (+1.8)	<u>85.29</u> (+0.0)	<u>13.33</u> (+3.3)
	Infix	**75.20** (+2.2)	**85.75** (+0.5)	6.67 (−3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	**20.00** (+10.0)
Base Models (Raw Text)				
Qwen2.5-Math-7B	Baseline	**72.20**	<u>84.15</u>	6.67
	Prefix	<u>71.60</u> (−0.6)	**84.31** (+0.2)	**16.67** (+10.0)
	Infix	63.20 (−9.0)	64.29 (−19.9)	**16.67** (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	<u>13.33</u> (+6.7)
Qwen2.5-Math-1.5B	Baseline	<u>61.40</u>	**77.86**	**16.67**
	Prefix	**64.40** (+3.0)	<u>74.30</u> (−3.6)	<u>10.00</u> (−6.7)
	Infix	63.20 (+1.8)	73.92 (−3.9)	<u>10.00</u> (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
Base Models + ChatML (Format Mismatch)				
Qwen2.5-Math-7B	Baseline	52.20	58.30	**23.33**
	Prefix	59.20 (+7.0)	56.41 (−1.9)	<u>3.33</u> (−20.0)
	Infix	<u>62.00</u> (+9.8)	<u>69.07</u> (+10.8)	**23.33** (+0.0)
	Suffix	**70.40** (+18.2)	**75.97** (+17.7)	**23.33** (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	<u>13.33</u>
	Prefix	**63.40** (+29.0)	**67.02** (+29.6)	**20.00** (+6.7)
	Infix	39.20 (+4.8)	50.42 (+13.0)	6.67 (−6.7)
	Suffix	<u>51.00</u> (+16.6)	<u>50.64</u> (+13.2)	<u>13.33</u> (+0.0)

Table 7: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

A.1 RSP Injection Positions and Prompt Templates

Below we show the prompt structure for each injection position across all model types. Blue text indicates RSP embeddings, and purple text indicates special tokens.

A.1.1 Prefix

RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

ChatML (Qwen Instruct / Base + ChatML).

```
[Random Embeddings (L tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
[Random Embeddings (L tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
[Random Embeddings (L tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.2 Infix

A description text and special tokens are inserted into the prompt. The special token positions are then replaced with RSP embeddings at the embedding level.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens: <special_token_1>...<special_token_L>
<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
```

```
Here are the {L} special tokens:
<reserved_special_token_0>...
<reserved_special_token_L>
<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning.  If these tokens are useful for your case, you can
use them as reference.  If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens:  <special_token_1>...<special_token_L>
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.3 Suffix

For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token. For raw text models, RSP embeddings are concatenated at the end of the prompt.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}[Random Embeddings (L tokens)]<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date:  December 2023
Today Date:  26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}[Random Embeddings (L tokens)]<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
Please reason step by step, and put your final answer within \boxed{}.[Random
Embeddings (L tokens)]
```

A.2 Answer Extraction and Grading

The final answer is extracted from the model output through a fallback chain. Each step is attempted in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. `"final answer is $...$.  I hope"` pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following `"the answer is"`
4. Text following `"final answer is"`
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing `$`, `\left`, `\right`, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose`, `rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature $= 0$) and our unified answer extraction pipeline.

TTSV. *Prefix* length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 8 reports the per-benchmark hyperparameters.

Table 8: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 6 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy, and Table 9 reports the corresponding scalar statistics including overall means, first-5% means, and the average number of generated tokens per problem. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

Table 9: Per-step statistics for Qwen2.5-Math-1.5B-Instruct on MATH-500 across the full generation and the first 5% of generation steps, together with the average number of generated tokens per problem. Top-1 Prob (overall) is reported as a weighted average over the first-10%/middle/last-10% segment means with weights $0.1/0.8/0.1$. Extension of Figure 3 (main text) and Figure 6.

Metric	Baseline	*Prefix*	*Infix*	*Suffix*	LTPO	SoftCoT	TTSV
Tokens (mean)	575.7	558.4	549.9	573.3	592.3	594.4	577.4
Entropy (first 5%)	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Entropy (overall)	0.0871	0.0881	0.0899	0.1049	0.0909	0.1059	0.0864
Top-1 Prob (first 5%)	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Top-1 Prob (overall)	0.9684	0.9681	0.9678	0.9647	0.9683	0.9651	0.9685
Varentropy (first 5%)	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174
Varentropy (overall)	0.1353	0.1375	0.1422	0.2038	0.1452	0.2404	0.1361

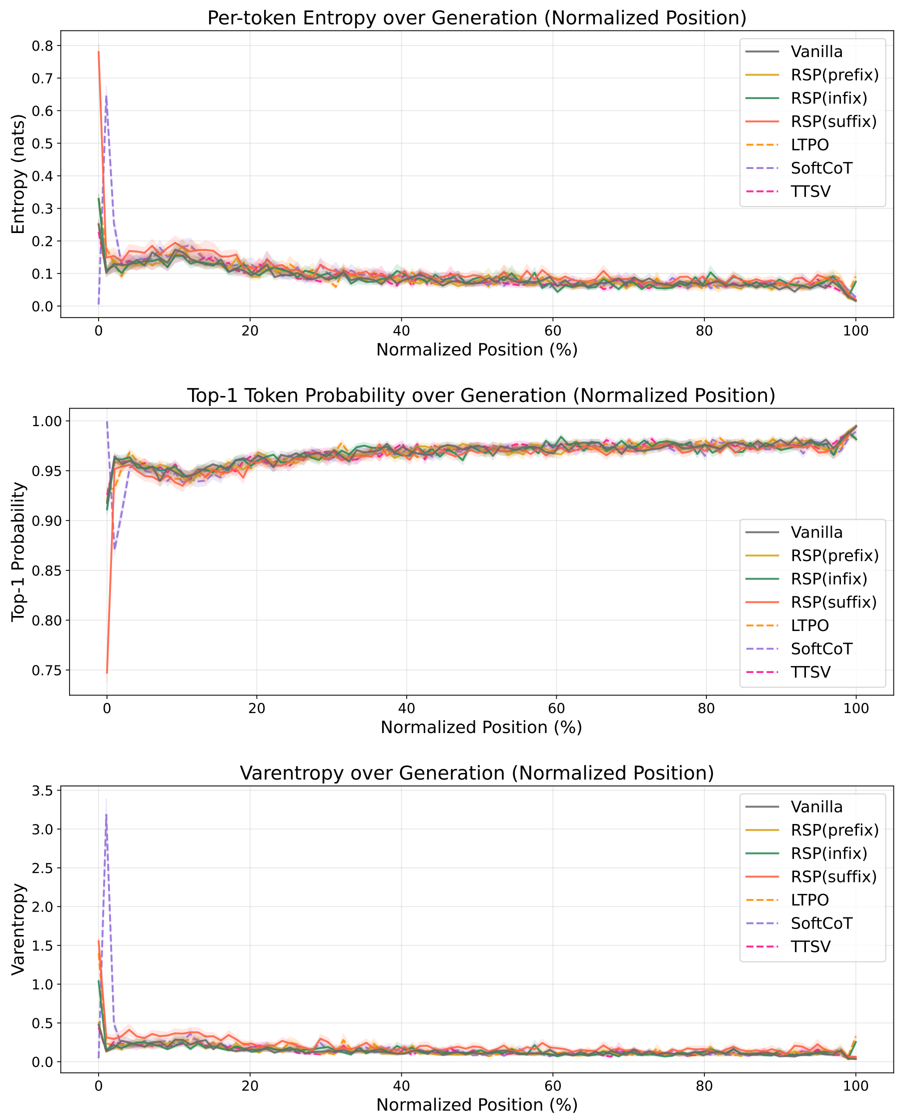


Figure 6: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior *soft prompt* methods. All methods converge after the initial stage.

B Token-Level Distribution Metrics

This appendix provides formal definitions for the metrics used in Section 4.4. Numerical results are reported in Table 3 of the main text. We extract first-token logits via a single forward pass (no autoregressive generation) and store the top-100 token ids and logit values per problem. Temperature conditions reuse the baseline logits scaled by $1/\tau$, requiring no separate inference. RSP draws 16 independent seeds per problem (MATH-500), and reported metrics are the mean across the 16 seeds. Pass@1 accuracy is computed over 16 samples per problem, with the baseline at $\tau{=}1.0$ reaching $73.92 \pm 0.78\%$. Let P_{base} denote the baseline next-token distribution at $\tau{=}1.0$ and P_{target} the candidate distribution being compared (e.g., baselines at $\tau{=}2.0, 3.0$ or RSP at $\tau{=}1.0$). All metrics are computed analytically from saved logits at the first generation step.

Spearman rank correlation. Spearman's ρ is the Pearson correlation coefficient between the token ranks of two distributions:

$$\rho \;=\; \mathrm{Pearson}(\mathrm{rank}(P_{\text{target}}),\ \mathrm{rank}(P_{\text{base}}))\,. \tag{3}$$

Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore indicates rank reorganization beyond what temperature scaling can produce.

Mass outside top-K. Let $\mathrm{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under P_{base}. The probability mass that the candidate distribution places *outside* this set is

$$\mathrm{MassOutside}_K \;=\; 1 - \sum_{t \in \mathrm{TopK}(P_{\text{base}})} P_{\text{target}}(t). \tag{4}$$

This directly measures support expansion: how much probability the candidate assigns to tokens that are not in the baseline's preferred set. Reported values use $K = 10$.

Jensen–Shannon divergence. The symmetrized, bounded version of Kullback–Leibler divergence:

$$\mathrm{JS}(P, Q) \;=\; \tfrac{1}{2}\,\mathrm{KL}(P \,\|\, M) + \tfrac{1}{2}\,\mathrm{KL}(Q \,\|\, M), \qquad M = \tfrac{1}{2}(P + Q). \tag{5}$$

We use base-2 logarithms so that $\mathrm{JS} \in [0, 1]$. Tokens absent from the stored top-100 are assigned a sentinel logit (-10^9) before softmax, which is negligible for our setting.

C Prompt-Law Properties Used by the Theory

This appendix isolates the only properties of the RSP law used in the main text: centeredness, direction-agnostic covariance under a variance budget, and positive probability on every nonempty open set. We avoid stronger semantic claims because Section 3.2 needs only these geometric and measure-theoretic facts.

C.1 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let z_a^{Base} denote the no-RSP output logit at vocab token a and $z_a^{\text{RSP}}(\bar{h})$ the corresponding logit when the centered RSP perturbation takes value $\bar{h}$. The (true) vocab-logit gap under RSP is $\Delta_{ab}(\bar{h}) := z_a^{\text{RSP}}(\bar{h}) - z_b^{\text{RSP}}(\bar{h})$, distinct from the no-RSP baseline gap $\Delta_{ab}^{\text{Base}} := z_a^{\text{Base}} - z_b^{\text{Base}}$. Note that $\bar{h} = 0$ corresponds to the centered RSP at its entrywise mean rather than to the no-RSP forward pass, so in general $\Delta_{ab}(0) \neq \Delta_{ab}^{\text{Base}}$. The transformer's logit map is non-linear in $\bar{h}$, but a first-order Taylor expansion around $\bar{h} = 0$ yields the local surrogate

$$\Delta_{ab}^{\text{lin}}(\bar{h}) \;:=\; \Delta_{ab} + b_{ab}^{\top}\bar{h},$$

with $\Delta_{ab} := \Delta_{ab}(0)$ and $b_{ab} := \nabla_{\bar{h}}(z_a^{\text{RSP}} - z_b^{\text{RSP}})|_{\bar{h}=0}$. The proposition below is stated for the surrogate Δ_{ab}^{lin}. For the true non-linear gap $\Delta_{ab}(\bar{h})$, the first-order term has zero mean under centered perturbations, but the Taylor remainder $r_{ab}(\bar{h}) = O(\|\bar{h}\|^2)$ may contribute a second-order mean shift; we make no claim about its sign or magnitude.

Proposition 2 (No systematic first-order drift in the surrogate)**.** *If* $\mathbb{E}[\bar{h}] = 0$*, then*

$$\mathbb{E}\big[\Delta_{ab}^{\text{lin}}(\bar{h})\big] = \Delta_{ab}$$

for every token pair (a, b)*. If a deterministic centered prompt* $\bar{h}_0$ *is reused across runs, then*

$$\Delta_{ab}^{\text{lin}}(\bar{h}_0) - \Delta_{ab} = b_{ab}^{\top}\bar{h}_0$$

is a fixed directional bias.

Proof. The centered case is immediate from linearity of expectation:

$$\mathbb{E}\big[\Delta_{ab}^{\text{lin}}(\bar{h})\big] = \Delta_{ab} + b_{ab}^{\top}\mathbb{E}[\bar{h}] = \Delta_{ab}.$$

The deterministic case follows by substitution. □

This proposition rules out a run-averaged first-order bias. It does not say that individual prompt draws leave the logits unchanged.

C.2 Proof of Proposition 1

For a unit vector u, the first-order perturbation energy available along u is $\text{Var}_{h\sim D}(u^{\top}h) = u^{\top}\Sigma_D u$. The worst-case directional energy is therefore the minimum Rayleigh quotient of Σ_D.

Proof. For any symmetric covariance matrix Σ_D,

$$\min_{\|u\|=1} u^{\top}\Sigma_D u = \lambda_{\min}(\Sigma_D).$$

Let the eigenvalues of Σ_D be $\lambda_1, \ldots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \leq \frac{1}{d}\sum_{m=1}^{d}\lambda_m = \frac{\text{tr}(\Sigma_D)}{d} \leq \frac{\rho^2}{d}.$$

Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. □

The proof uses only covariance. We choose a Gaussian law for RSP because it adds the open-set reachability property proved next.

C.3 Open-Set Reachability of Isotropic Gaussian Prompts

Proposition 3 (Open-set reachability). *Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ with $\sigma > 0$. Then every nonempty open set $U \subseteq \mathbb{R}^d$ satisfies*

$$\Pr(\bar{h} \in U) > 0.$$

Proof. The Gaussian density

$$(2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|\bar{h}\|^2}{2\sigma^2}\right)$$

is strictly positive at every point of $\mathbb{R}^d$. Every nonempty open set contains a ball of positive Lebesgue measure, and integrating a strictly positive density over that ball gives positive probability. $\square$

This is the only support property used in the informal branching argument of §3.2 and in Appendix D.3.3.

D Proofs for the Transformer-Level Mechanism

This appendix follows the same order as the main theory section: exploration, annealing, and performance gain. The prompt-law facts used before these proofs are collected in Appendix C.

D.1 Exploration: attention decomposition and perturbation size

Proposition 4 (Attention decomposition)**.** *For one attention head at query position i in layer ℓ, let $\alpha_{ij}^{(\ell)}$ be the attention weights over $n \geq 1$ unmasked real tokens and $L \geq 1$ random tokens, with finite attention logits. Define*

$$w_{r,i}^{(\ell)} := \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)}$$

and, for $j \in [n]$,

$$\tilde{\alpha}_{ij}^{(\ell)} := \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}}.$$

Then

$$\mathbf{o}_i^{(\ell)} = (1 - w_{r,i}^{(\ell)})\, \tilde{\mathbf{o}}_i^{(\ell)} + \boldsymbol{\eta}_i^{(\ell)},$$

where

$$\tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^{n} \tilde{\alpha}_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} \qquad \textit{and} \qquad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

Proof. The head output is

$$\mathbf{o}_i^{(\ell)} = \sum_{j=1}^{n} \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

By the softmax positivity and the existence of at least one unmasked real token, $\sum_{j=1}^{n} \alpha_{ij}^{(\ell)} = 1 - w_{r,i}^{(\ell)} > 0$, so

$$\sum_{j=1}^{n} \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \sum_{j=1}^{n} \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)})\, \tilde{\mathbf{o}}_i^{(\ell)}.$$

Substituting this identity gives the decomposition. □

Corollary 1 (Magnitude scales with random-token attention)**.** *For every realization of the random values,*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

Proof. By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(\ell)}\| = \left\| \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)} \right\| \leq \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \|\mathbf{v}_{r_j}^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

□

Corollary 1 is the only quantitative fact needed in the main text. No independence assumption between attention weights and random keys is required. Once $w_{r,i}^{(\ell)}$ is small, the random contribution must be small as well.

D.2 Corollaries of Theorem 1 on KV cache growth

The fixed-gap decay bound yields the corollaries below.

Corollary 2 (Sequence-wise annealing of the fixed-gap envelope)**.** *For fixed L and $\Delta_i^{(\ell)}$, the upper bound*

$$f(n) = \frac{L}{L + n\exp(\Delta_i^{(\ell)}/\sqrt{d_{head}})}$$

is strictly decreasing in n and tends to 0 as $n \to \infty$. KV cache growth alone therefore narrows the largest RSP attention mass compatible with that gap.

During autoregressive decoding $\Delta_i^{(\ell)}$ also moves because queries, keys, and hidden states co-evolve. The accurate reading is not "the realized mass decreases at every step" but "the envelope compatible with a fixed gap shrinks as the KV cache grows."

Proof. Differentiate $f(n)$ with respect to n:

$$f'(n) = -\frac{L\exp(\Delta_i^{(\ell)}/\sqrt{d_{\text{head}}})}{\Big(L + n\exp(\Delta_i^{(\ell)}/\sqrt{d_{\text{head}}})\Big)^2} < 0.$$

Thus $f(n)$ is strictly decreasing, and its denominator diverges linearly in n while its numerator is fixed, so $f(n) \to 0$. $\square$

Corollary 3 (Vanishing random contribution under annealing)**.** *If $n \to \infty$ while L and $\Delta_i^{(\ell)}$ remain fixed, $w_{r,i}^{(\ell)} \to 0$, and for every realization of the random values*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \le w_{r,i}^{(\ell)} \max_{j\in[L]} \|\mathbf{v}_{r_j}^{(\ell)}\| \to 0.$$

Proof. $w_{r,i}^{(\ell)} \to 0$ by Corollary 2. Apply Corollary 1 to bound $\|\boldsymbol{\eta}_i^{(\ell)}\|$. The maximum norm is finite for any realization of finitely many sampled values. $\square$

D.3 Performance gain: from local admission to Pass@N

From here we work at the vocab-logit and task levels, dropping the per-layer index ℓ. The main text uses a first-order surrogate to state two claims: the §3.2 claim that a baseline-excluded vocab token can enter the local top-K set, and the §3.3 closing that independent reruns turn such local openings into Pass@N gains. The proofs below use only one support condition on the centered prompt law D:

$$\text{(S1)} \qquad D(U) > 0 \quad \text{for every nonempty open set } U \subseteq \mathbb{R}^d.$$

Proposition 3 shows that the isotropic Gaussian law of §3.2 satisfies (S1).

D.3.1 Autoregressive Formulation

In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^{T} p(y_t \mid x, y_{<t}),$$

where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single global distribution.

D.3.2 Local Linearization of Logits under RSP

Throughout this subsection, $\bar{h} \in \mathbb{R}^d$ denotes *one* centered prompt vector from the RSP definition (Eq. (1) in §3.1), treated as a per-token surrogate. The actual implementation uses a length-L prompt $\mathbf{H} = (h_1, \ldots, h_L)$ with L independent draws; the arguments below isolate how one local perturbation can shift the logits. We model the perturbed logits as a function $z_a(\bar{h})$ at vocab token a, assuming local smoothness in a neighborhood of $\bar{h} = 0$:

$$z_a(\bar{h}) = z_a(0) + \nabla_{\bar{h}} z_a(0)^\top \bar{h} + r_a(\bar{h}),$$

where $r_a(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $c_a := \nabla_{\bar{h}} z_a(0)$ and $z_a := z_a(0)$, we obtain the local affine surrogate

$$z_a^{\mathrm{lin}}(\bar{h}) := z_a + c_a^\top \bar{h}.$$

This approximation is used as a first-order surrogate for branch opening, not as an exact global model of the transformer logits.

D.3.3 Top-K entry region (Remark)

This complements the informal branching argument in §3.2. Define the entry region

$$\mathcal{G}_{a,K} \ := \ \left\{ \bar{h} \in \mathbb{R}^d : \#\{ b \neq a : z_b^{\mathrm{lin}}(\bar{h}) > z_a^{\mathrm{lin}}(\bar{h}) \} \leq K - 1 \right\}.$$

If $\mathcal{G}_{a,K}$ contains a nonempty open set U, then by (S1) $D(U) > 0$, and $U \subseteq \mathcal{G}_{a,K}$ gives $\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{a,K}) \geq D(U) > 0$. A useful sufficient condition is the existence of some $\bar{h}_0$ at which token a strictly outranks all but at most $K-1$ other tokens in the affine surrogate; by continuity, a neighborhood of $\bar{h}_0$ is then contained in $\mathcal{G}_{a,K}$. This is a rank-changing event unreachable by the monotone temperature transform, but it does not by itself imply (M1).

D.3.4 Pass@N ensemble bound (Remark)

The bound $\Pr(\mathrm{Pass@}N \text{ on } x) \geq 1 - (1 - p_{\min}(x))^N$ in the main text is a standard independent-Bernoulli union argument. Let A_n be the event that the n-th run reaches a correct trajectory on x; under (M1) $\Pr(A_n) \geq p_{\min}(x)$, and under (M2)

$$\Pr\Big(\bigcap_{n=1}^{N} A_n^c \Big) \leq (1 - p_{\min}(x))^N, \qquad \Pr\Big(\bigcup_{n=1}^{N} A_n \Big) \geq 1 - (1 - p_{\min}(x))^N \ \geq \ 1 - e^{-N p_{\min}(x)}.$$

This is a generic fact, not specific to RSP; RSP's role is to make branch opening possible and to support (M2) through independent resampling, while whether a branch satisfies the task-side correctness condition (M1) remains task-dependent.

E Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 2 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

E.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,m)}$ denote the post-softmax attention weight at layer ℓ, head index m (using m to avoid collision with the vocab token a of §3.2 and the RSP matrix $\mathbf{H}$), from query position i to key position j, satisfying $\sum_j \alpha_{i,j}^{(\ell,m)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let N_{head} be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\mathrm{PTM}_Q[\ell, i] = \frac{1}{|Q|\, N_{\text{head}}} \sum_{m=1}^{N_{\text{head}}} \sum_{j \in Q} \alpha_{i,j}^{(\ell,m)}, \qquad \mathrm{PTM}_R[\ell, i] = \frac{1}{|R|\, N_{\text{head}}} \sum_{m=1}^{N_{\text{head}}} \sum_{j \in R} \alpha_{i,j}^{(\ell,m)}. \quad (6)$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

E.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_\bullet^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \mathrm{PTM}_\bullet[\ell, i]. \quad (7)$$

Figure 2 reports the sample average of Eq. (7) over the 500 valid samples.

E.3 Excluding the `\boxed{...}` Span

The reasoning position range terminates strictly before the final `\boxed{...}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{...}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

E.4 Excluding the Final Two Layers

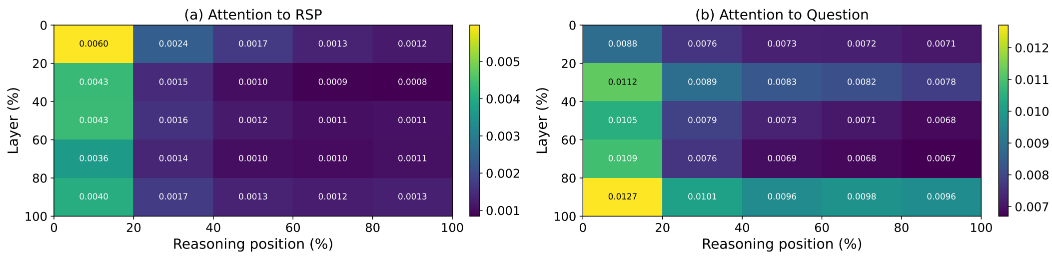
The layer axis further excludes the last two transformer layers. This choice is motivated by a standard late-layer effect rather than by any model-specific tuning decision.

Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

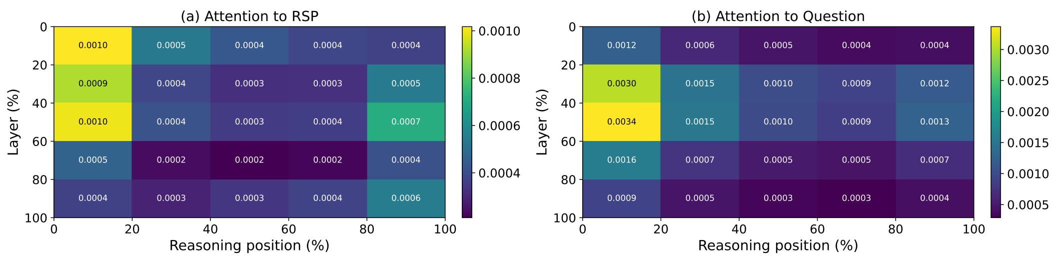
Excluding the final two layers keeps the heatmap focused on reasoning-phase dynamics instead of readout-phase dynamics. This exclusion is not load-bearing for the main claim: the sequence-direction attenuation emphasized in the main text is also visible without it, and Theorem 1 concerns sequence-direction attenuation rather than layer-wise monotonicity.

E.5 Additional Suffix Heatmaps

For the late-layer reasons discussed above, the pattern in Figure 7 does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across both models: question-token attention varies broadly across layers, whereas RSP attention shows sequence-wise decay consistent with Theorem 1 alongside an empirical layer-wise attenuation that the theorem itself does not predict.



(a) Qwen2.5-Math-1.5B-Instruct, *suffix*



(b) LLaMA-3.1-8B-Instruct, *suffix*

Figure 7: Per-token RSP attention mass under *suffix* injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 2: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

F DAPO Implementation

This appendix complements Section 4.5 with the DAPO loss modifications relative to GRPO, the DAPO + RSP implementation, and full per-step results.

F.1 From GRPO to DAPO

The vanilla GRPO objective [Shao et al., 2024] for a group of G rollouts $\{y_i\}_{i=1}^{G}$ (each $y_i = (y_{i,1}, \ldots, y_{i,|y_i|})$ a token sequence; we use y rather than o throughout this appendix to avoid collision with the head output $\mathbf{o}_i^{(\ell)}$ of §3.2) with rewards $\{R_i\}_{i=1}^{G}$ is

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|y_i|}\sum_{t=1}^{|y_i|}\Big(\min\big(r_{i,t}\hat{A}_{i,t},\, \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon)\hat{A}_{i,t}\big) - \beta\, D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})\Big)\right], \tag{8}$$

with importance ratio $r_{i,t} = \pi_\theta(y_{i,t} \mid q, y_{i,<t})/\pi_{\theta_{\text{old}}}(y_{i,t} \mid q, y_{i,<t})$, group-relative advantage $\hat{A}_{i,t} = (R_i - \text{mean}(\{R_i\}))/\text{std}(\{R_i\})$, clipping range ε, and KL penalty $\beta\, D_{\text{KL}}$ against the reference π_{ref}.

We build on the SimpleRL-Zoo [Zeng et al., 2025] training pipeline and implement DAPO on top of VeRL [Sheng et al., 2025]. DAPO modifies Eq. (8) in four ways: (i) token-level loss aggregation $1/\sum_i |y_i|$ instead of $1/|y_i|$, (ii) asymmetric clipping with $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}}) = (0.2, 0.28)$, (iii) an implementation-side negative-advantage dual-clipping safeguard $\max(L_{\text{clipped}}, c\hat{A})$ with $c = 10$ (active only when $\hat{A} < 0$; omitted from Eq. (9) below for readability), and (iv) KL terms removed. The resulting DAPO objective is

$$\mathcal{J}_{\text{DAPO}}(\theta) \;=\; \mathbb{E}\left[\frac{1}{\sum_i |y_i|}\sum_{i,t}\min\Big(r_{i,t}\hat{A}_{i,t},\, \text{clip}\big(r_{i,t},\, 1-\varepsilon_{\text{low}},\, 1+\varepsilon_{\text{high}}\big)\hat{A}_{i,t}\Big)\right]. \tag{9}$$

F.2 DAPO + RSP Implementation

For DAPO + RSP, we additionally inject a per-rollout RSP $\mathbf{H}_i \in \mathbb{R}^{L\times d}$ ($L = 20$) into each rollout. Each $\mathbf{H}_i$ is generated deterministically from a per-step rollout seed and injected via a forward hook; it is not a trainable parameter, so the learning signal is the policy adapting to arbitrary $\mathbf{H}_i$ rather than a learned prompt. Including $\mathbf{H}_i$ in the log-probabilities at both rollout and update time keeps the update on-policy with respect to the RSP-conditioned distribution (avoiding off-policy mismatch). With $G = 8$ rollouts per prompt, each rollout receives an independently drawn $\mathbf{H}_i$.

F.3 Data, Batch, and Hyperparameters

Table 10: DAPO / DAPO + RSP training setup on Qwen2.5-Math-7B.

Field	Value
Train data	MATH Level 3–5 subset (`simplelr_math_35`, ~8,500 prompts)
Prompt template	`qwen-boxed`
Train batch size	1024
PPO mini-batch size	256 (4 PPO updates per rollout)
Rollouts per prompt G	8
Max prompt / response length	2,028 / 2,048
Rollout sampling	temperature 1.0, top-p 1.0, top-k 50
$(\varepsilon_{\text{low}}, \varepsilon_{\text{high}})$	$(0.2,\ 0.28)$
Dual clip c	10.0
Loss aggregation	token-mean $(1/\sum_i \lvert y_i \rvert)$
KL terms	disabled
Entropy coefficient	0
Optimizer	AdamW, learning rate 5×10^{-7} (constant), no warmup
Total training	~10 epochs, ≥ 80 optimization steps
Save / eval frequency	every 10 steps
RSP (DAPO + RSP only)	*suffix* injection, $L = 20$, freshly drawn per rollout
Hardware	$4\times$ NVIDIA B200 GPUs
Wall-clock training time	$\sim$12 hours per run

F.4 Evaluation Protocol

Each saved checkpoint is converted to HuggingFace format and evaluated using the SimpleRL-Zoo `eval_math_nodes.sh` pipeline, which selects sampling parameters per benchmark to reduce variance on the smaller competition sets:

Table 11: Per-benchmark evaluation protocol used by SimpleRL-Zoo `sh/eval.sh`.

Benchmark	# Problems	Temperature	n_{sampling}	Metric
AIME 2024	30	1.0	32	Avg@32
AMC 2023	40	1.0	32	Avg@32
MATH-500	500	0.0	1	accuracy (mean@1)
GSM8K	1,319	0.0	1	accuracy
OlympiadBench	675	0.0	1	accuracy
Minerva Math	272	0.0	1	accuracy
GaoKao 2023-EN	385	0.0	1	accuracy

The maximum number of generated tokens is $16{,}000$ across all benchmarks. The reported average is the unweighted mean over the five benchmarks (GSM8K, MATH-500, College Math, Minerva Math, AIME 2024).

F.5 Per-Step Average Accuracy

Table 12: Five-benchmark average accuracy (%) at every checkpoint, computed over GSM8K, MATH-500, College Math, Minerva Math, and AIME 2024. **Bold** marks each method's peak.

Step	DAPO	DAPO + RSP
0	30.06	30.06
10	46.40	46.22
20	49.02	49.54
30	49.58	50.72
40	50.62	52.06
50	50.88	52.40
60	51.84	52.96
70	52.52	53.86
80	**53.20**	54.04
90	52.52	**54.36**
100	50.54	53.64

G Broader Impacts

This work provides empirical and theoretical analysis of how random embedding injection affects the reasoning behavior of large language models. The contributions are primarily foundational, with potential downstream implications for reinforcement learning-based post-training of reasoning models such as GRPO.

Positive impacts. RSP-induced trajectory diversity could improve the sample efficiency of RL-based LLM post-training by providing richer learning signals within group-relative advantage estimation. This has the potential to reduce compute costs associated with alignment and reasoning-oriented fine-tuning, making such methods more accessible to research groups with limited resources.

Potential concerns. Methods that expand the effective output support of LLMs may increase the reachability of low-probability tokens, which includes both desirable alternative reasoning paths and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems should combine it with existing safety filtering and alignment mechanisms rather than treating it as a standalone diversity source.

Limitations on impact scope. Since this work focuses on analysis rather than deploying a new model or dataset, the direct societal impact is limited. The broader implications described above are contingent on future work integrating RSP into applied training pipelines.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and §1 state four scoped claims, each tied to a section: (i) RSP's contribution at each attention layer is captured by a single scalar with a fixed-gap upper-envelope cache-growth bound (Theorem 1, §3.3); (ii) under a local affine surrogate, isotropic re-sampling reaches top-K entry regions unattainable by rank-preserving temperature sampling (§3.2); (iii) RSP raises early-token entropy and stabilizes later, with Pass@N gains under independent re-sampling (§4); and (iv) the diversity transfers to RL training when combined with DAPO (§4.5). We do not claim a theorem that directly explains the format-mismatch recovery (Table 1); the wrong-attractor hypothesis there is explicitly informal.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Limitations are noted in several places rather than in a dedicated section: (a) the multi-path Pass@N argument requires the task-side assumption $p_{\min}(x) > 0$, flagged in §3.3; (b) the +18–29 pp recovery on the format-mismatch rows of Table 1 is empirically observed but not formally derived from any theorem in §3, with the EOS-derailment hypothesis stated informally in §4.2 and its quantitative verification left for future work; (c) Table 1 reports per-position accuracy across {Prefix, Suffix}, with per-position variance large and several cases below baseline, so position is treated as a hyperparameter, and the RSP length L is selected per model from $\{10, 15, 20\}$ on the evaluation set with held-out validation left for future work (§5); (d) Theorem 1 guarantees only sequence-axis attenuation, while the observed layer-axis attenuation is an empirical phenomenon from deep-layer attention-logit gap sharpening that lies outside the theorem's envelope (§5); (e) the results are confined to RoPE-based decoders and math reasoning, with generalization to other position encodings and domains left for future work (§5). The future-work items in §5, namely generalization beyond RoPE/math reasoning, deep-layer sharpening, and held-out hyperparameter selection, correspond respectively to (e), (d), and (c).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Theorem 1 (cache-growth attenuation) carries an inline proof in §3.3. Proposition 1 is stated in §3.2 with its proof in Appendix C; the supporting top-K entry and Pass@N ensemble arguments around it are presented as informal discussion in §3.2 with formal write-ups in the appendix. Each statement explicitly lists its assumptions: finite logits and $n, L \geq 1$ for Theorem 1; finite second-moment budget for Proposition 1; and the locally affine surrogate plus $p_{\min}(x) > 0$ task-side condition together with mutual independence of the N runs for the Pass@N argument.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: §4.1 and Appendix A specify benchmarks (MATH-500, GSM8K, AIME 2024 in the main table; further sets used for the DAPO study), models (LLaMA-3.1-8B-Instruct and Qwen2.5-Math 1.5B/7B variants), the RSP injection grid (Prefix/Infix/Suffix; $L \in \{10, 15, 20\}$), greedy decoding, and the answer extraction pipeline. Appendix F provides the full DAPO + RSP training recipe (batch of 1,024 prompts, $G = 8$ rollouts per prompt at $\tau = 1.0$, four PPO mini-batch updates of 256 prompts each, 100 training steps, suffix RSP $L = 20$ during rollouts only).

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: All datasets and base models used in the paper are publicly available and cited (MATH-500, GSM8K, AIME 2024, AMC 2023, OlympiadBench, Minerva Math, GaoKao 2023-EN, College Math; Qwen2.5-Math 1.5B/7B and LLaMA-3.1-8B-Instruct). RSP itself is described to a level of detail sufficient for re-implementation (Eq. (1) and §4.1). However, our RSP injection harness and DAPO + RSP training pipeline are not yet released; we will release an anonymized implementation alongside the camera-ready submission.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: §4.1 and Appendix A list all benchmarks, models, RSP injection settings (positions, lengths, per-row L choice in Appendix Table 7), maximum generation length per benchmark, and decoding strategy. Appendix F provides the full DAPO + RSP training settings, including optimizer, batch composition, mini-batch updates, asymmetric clipping range $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}}) = (0.2, 0.28)$, dual clipping safeguard, and RSP injection protocol during rollouts.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [No]

   Justification: Pass@N experiments use $N = 16$ samples per problem (§4.4), Table 3 reports mean$\pm$standard deviation over 16 RSP seeds, and AIME 2024 is scored by Avg@32 to reduce variance. However, Table 1 (Table 1) reports single-run accuracies without bootstrap confidence intervals or paired tests, and the DAPO + RSP training in §4.5 uses a single training seed. Multi-seed evaluation, particularly on small competition sets (AIME 2024, AMC 2023) where variance is highest, is left for future work.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Inference experiments (Tables 1, 2; Figure 3) run on a single NVIDIA A6000 40GB GPU with greedy decoding (Appendix A). DAPO + RSP training (§4.5) runs on $4\times$ NVIDIA B200 GPUs for approximately 12 hours per run (Appendix F, hardware row of the configuration table).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work uses publicly available math reasoning benchmarks and pretrained language models, involves no human subjects or personal data, and does not introduce a release that bypasses standard model-safety considerations. We have reviewed the NeurIPS Code of Ethics and identify no conflicts.

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Appendix G discusses both directions: positive impact via more sample-efficient RL post-training of reasoning LLMs, and a potential downside that broadening the effective output support also increases reachability of undesirable trajectories, which we argue should keep RSP coupled with existing safety filtering rather than replacing it.

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release new pretrained models, generative systems, or scraped datasets. RSP is a generation-time perturbation applied to existing pretrained models and introduces no separate artifact requiring safeguards.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All datasets (MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], AIME 2024, AMC 2023, OlympiadBench [He et al., 2024], Minerva Math [Lewkowycz et al., 2022], GaoKao 2023-EN, College Math) and pretrained models (Qwen2.5-Math [Yang et al., 2024], LLaMA-3.1 [Grattafiori et al., 2024]) are publicly available and cited at first use. The training pipeline builds on SimpleRL-Zoo [Zeng et al., 2025] and VeRL [Sheng et al., 2025], also cited. Each asset is used under its original publicly stated terms.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: The paper does not release new datasets or pretrained models. The implementation code planned for camera-ready release (item 5) will be accompanied by configuration files and a README; no derived dataset or model checkpoint is part of the submission.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper involves no crowdsourcing and no research with human subjects. All evaluations use static, publicly available math reasoning benchmarks.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: As noted in item 14, the paper involves no human subjects, so IRB review is not applicable.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: Pretrained LLMs (Qwen2.5-Math, LLaMA-3.1) appear in the paper as the experimental subject of analysis, not as a non-standard component of the methods themselves. The methods (RSP definition, attention decomposition, DAPO + RSP training) do not rely on auxiliary LLMs in any non-standard or originality-affecting role.